



6406 - Unlocking High-Angular Resolution Spectroscopy for JWST using Kernel Phase Interferometry and the NIRSpec IFU

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

This proposal will develop a kernel phase interferometry (KPI) software pipeline for the NIRSpec IFU mode. KPI is a data analysis technique which effectively doubles the angular resolution at which companions (or other asymmetries) can be resolved in images taken from space or behind ground-based extreme adaptive optics systems. The kernel phase is an interferometric observable constructed by using the phase information in a high-Strehl image, boosting the resolution to double that of the classical diffraction limit. In the context of NIRSpec, KPI can unlock high and moderate resolution spectroscopy at smaller angular separations than what can be achieved with other JWST instruments. This opens up NIRSpec to a wealth of observations of young protoplanets, bright exoplanets and other substellar objects at small orbital periods. This is particularly important as we know that the peak occurrence rate of Jovian mass planets are at ~ 3 AU and GAIA astrometry, next generation direct imaging missions and the ELTs are set to discover planets at these separations. It is therefore critical to develop a NIRSpec IFU KPI pipeline to bolster spectroscopic follow-up capabilities at small angular separations. This would allow characterisation of these kinds of systems at longer wavelengths which are only accessible from space with the JWST.

OBSERVING DESCRIPTION

This proposal is an Archival Go Software proposal to develop a kernel phase interferometry (KPI) pipeline for the NIRSpec instrument when using the IFU mode. The proposal will make use of available NIRSpec archival data for testing and characterisation of the performance of the software. KPI is not restricted to a particular filter or grism setting. Preliminary analysis will be conducted with the VHS J1256–1257 b data from the JWST Early Release Science Programme.